Yongjian Huang （黄永健）

Birth	2002.01.27	**Hometown**	Guangdong, China
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Res. Int.

Operations Research: Assignment problems, optimization algorithms;

Artificial Intelligence: Machine learning, deep learning.

Education

September 2023 – March 2026, Shanghai University

Management Science and Engineering (Big Data and Business Artificial Intelligence track), Master, Supervisor: Prof. Jian Zhou, GPA: 3.7/4, Average score: 86.47/100

September 2019 – July 2023, University of South China

Business Administration (Enterprise Management track),
Bachelor, Supervisor: Prof. Tian Xie, GPA: 3.33/5, Average score: 83.87/100

Work Exp.

August 2024 – September 2025 Chinese Language Volunteer Teacher, Confucius Institute at University of Bahrain in Bahrain

- **One year** of international Chinese teaching experience in Bahrain.
- Taught and communicated with students in both **English** and **Arabic**.
- Organized and hosted the First Arabian Region Chinese Education Symposium 2024.
- Responsible for HSK examination administration.

November 2021 – June 2023, Student Union, Senior Member, Deputy Director, Director

- Demonstrated **strong hands-on skills** in problem-solving and process innovation.
- Developed Python-based **office automation software** to streamline task.
- Received recognition from the **college chairman and student advisors.**

Academic Exp.

Presided over the 2021 National College Student Innovation and Entrepreneurship Training Program Project (2021-2023), successfully concluded. *(Principal Investigator)*

- Participated in an undergraduate supervisor-led project funded by the National Natural Science Foundation of China, focusing on the *Cross-border parallel evolution mechanism and control strategies of nuclear-related network public opinion*, successfully concluded.
- Led a **national-level research project,** *Intelligent decision support system for student organizations*.
- Successfully applied for **seven computer software copyrights**, including one for an *intelligent decision support system for student work*, all of which have been implemented in actual student work at the undergraduate college.

Participated in the 2022 National College Student Innovation and Entrepreneurship Training Program Project (2022-2024), successfully concluded. *(Core Team Member)*

- Conducted a **national-level research project**, I*nvestigation and research on the construction of an industrial ecosystem based on e-commerce for poverty alleviation in the post-pandemic era, using Tuliu Group's digital initiatives for rural revitalization as a case study*.
- Led a project that won the **second prize** in the 9th "Challenge Cup" Extracurricular Academic Science and Technology Works Competition at the University of South China. *(Principal Investigator)*

Publications

[1] Xie, T.*, **Huang, Y. J.***, & Chen, W. F. **(2025)**. Multi-dimensional assignment model and its algorithm for multi-features decision-making problems. ***Expert Systems with Applications***, 270, 126369. DOI: 10.1016/j.eswa.2024.126369 (**Q1 TOP SCI, IF:7.5**)

[2] **Huang, Y. J.**, Huo, S.*, & Dong, Y. **(2025)**. Exploring the role of AI in smart grids: a 20-year bibliometric-based overview. ***International Journal of General Systems***, 1–32. DOI: 10.1080/03081079.2025.2547187 **(Q2 SCI)**

[3] **Huang, Y. J .**, Qader M. R., Zhou J*, Zhang X. **(2025)**. Prediction and path planning framework of X city's carbon emissions based on long short-term memory network model. ***Polish Journal of Environmental Studies***. Online first. DOI: 10.15244/pjoes/199609

[4] Xie, T., Dai, G. Y., Chen, W. F., Yang, C. P., **Huang, Y. J.**, & Wei, Y. Y.* **(2025)**. Pandemic Triggered Emergency Supply Chain Management Innovations: A Scientometric Analysis Based on Bibliometrics and Dynamic Topic Models. ***Disaster Medicine and Public Health Preparedness***, 19, e88. DOI:10.1017/dmp.2025.88

[5] Who Holds the Reins of Power? – Platforms Challenging Traditional Offline Transactions *(Working Papers)*

[6] Two-phase optimization methods for project scheduling problem with uncertain activity durations *(Working Papers)*

[7] Timely and equitably allocation of medical resources considering secondary mortality under public health emergency *(Working Papers)*

The abstracts and full texts of the publications can be viewed and downloaded from my homepage.

Honors

[1] Apr 2022 – May 2023 Obtained **7 software copyrights**, including the *Smart Decision Support System for Student in Universities*, among others.

[2] Dec 2023 – Second Prize (National Level), 20th "Huawei Cup" China Postgraduate **Mathematical** Contest in Modeling (Team Leader)

[3] May 2021 – Third Prize (National Level), 11th "CP Group Cup" National College Student **Market Research & Analysis** Competition (Core Leader)

[4] Sep 2021 – Third Prize (Provincial Level), 7th Hunan Province **"Internet+"** College Student Innovation & Entrepreneurship Competition (Deputy Leader)

[5] Apr 2021 – Second Prize (Provincial Level), 7th National College Student **Engineering Training** Comprehensive Ability Competition (Core Leader)

[6] May 2023 – Hunan Province 2023 Outstanding Graduates in **Innovation and Entrepreneurship**

[7] Apr 2024 – President's e-Learning Innovation Award, University of Bahrain, Bahrain

[8] Sep 2025 – "China-Bahrain Cultural Exchange Contributor", Embassy of the People's Republic of China in the Kingdom of Bahrain

[9] Sep 2025 – National Scholarship Award, PRC (2025)

Language

English (IELTS Overall Band Score: 6.5 , Writing band: 7.5); Mandarin (National Proficiency Test of Putonghua level 2 grade B); Arabic; Cantonese; Teochew

Skills

- **Python & Operations Research Modeling —** 6 years of hands-on experience in

developing optimization models and algorithms for resource allocation and scheduling problems, directly applied in published research (e.g., multi-dimensional assignment models in *Expert Systems with Applications*).

- **Web Scraping —** Proficient in using requests, BeautifulSoup, and Scrapy for large-scale data collection to support research data acquisition.
- **Automated Data Processing —** Designed Python scripts for batch processing of large datasets, significantly improving workflow efficiency in organizational operations; related work contributed to 7 registered software copyrights (e.g., Organization DSS).
- **Data Analysis & Visualization —** Experienced in using Pandas, NumPy, and Matplotlib to analyze research data and create visualizations for publications and competitions (e.g., "Huawei Cup" Mathematical Modeling Contest).
- **Computer Vision & Deep Learning —** Self-directed projects in image recognition and object detection using OpenCV, YOLO, PyTorch, and TensorFlow, with applications in face recognition and related tasks.
- **Statistical Analysis —** Skilled in applying SPSS and EViews for statistical analysis and econometric modeling, with practical experience from national-level research competitions.
- **Bibliometric Analysis —** Familiar with VOSviewer and CiteSpace for conducting bibliometric studies, including research trend mapping and citation network analysis, as utilized in published papers.
- **Scientific Visualization —** Proficient in creating diagrams with Visio and publication-quality charts with Matplotlib for academic papers and project reports.
- **Office & Typesetting Tools —** Skilled in LaTeX for typesetting, CorelDRAW for graphic design, and Word, Excel, PowerPoint for documentation and presentations.
- **Design & Media —** Experienced in image editing with Photoshop (PS) and video production with Premiere (PR), supporting content creation for academic and volunteer activities.

Self- Rec.

Who Am I: A Long-Term-Oriented Doer

I define myself as an optimistic and resilient individual who thrives on challenges and

long-term goals. Under the continuous guidance of **Prof. Tian Xie**, I spent four years from my undergraduate studies to complete and publish my first **Top SCI paper**. This journey strengthened my belief that meaningful research requires patience, persistence, and the courage to iterate repeatedly. For me, progress in science comes not from short bursts of effort but from sustained refinement and long-term dedication.

When Curiosity Led to Idea: From Concepts to Publications

My research journey began in **2020** during my sophomore year, when I first became involved in assignment problem solving under the mentorship of **Prof. Tian Xie**. Since then, I have navigated the entire research pipeline — from formulating research questions and identifying knowledge gaps, to designing experiments, writing manuscripts, submitting to journals, and responding to reviewer feedback. I have learned how to transform raw results into clear visualizations and compelling narratives, participated in the proposal preparation of National Natural Science Foundation of China (NSFC), MOE (Ministry of Education) Humanities and Social Sciences Research Project (Shanghai), and Shanghai Philosophy and Social Science Planning Project, and conducted peer reviews for SCI-indexed journals (Soft Computing etc.).

How I Conduct Research: Problem-Driven, Code-Enhanced

My research philosophy combines **problem-orientation** with **data-driven solutions**. During my undergraduate years, I developed a habit of identifying inefficiencies and designing highly efficient solutions — a mindset that now drives my academic work. In my graduate studies, I integrate Python, optimization algorithms, and computational modeling to tackle complex issues such as scheduling, and resource allocation. This ability to transform real-world scenarios into rigorous research questions allows me to consistently generate innovative insights across multiple domains.

Where I Grew: Cross-Cultural Experience in Bahrain

Thanks to the support of **Prof. Jian Zhou**, I spent a year living, studying and working at the *Confucius Institute of University of Bahrain*, where I managed to continue my research while expanding my experience in international education. Engaging with **Arabic-speaking students** and local institutions sharpened my ability to translate abstract methods into *practical solutions*, while also broadening my perspective on how institutional, cultural, and data environments influence problem-solving strategies. This experience nurtured an increasingly global perspective, allowing me to think beyond a single context and to identify insights from cross-regional comparisons.

What I Bring: Creativity and "Find-and-Fix" Capability

I bring a unique blend of **creativity** and **practical problem-solving skills**. Whether

automating data collection pipelines, streamlining workflows, or debugging processes, I continually refine systems through small, high-impact improvements. My goal is to ensure that my research not only advances theory but also generates solutions that work in real-world contexts. While I respect foundational theories, I am deeply passionate about applying them to address complex practical problems — because **I believe this is where research creates its greatest value**.